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Problem Solving Protocol

*m*CNN-ETC: identifying electron transporters and their functional families by using multiple windows scanning techniques in convolutional neural networks with evolutionary information of protein sequences

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Abstract

In the past decade, convolutional neural networks (CNNs) have been used as powerful tools by scientists to solve visual data tasks. However, many efforts of convolutional neural networks in solving protein function prediction and extracting useful information from protein sequences have certain limitations. In this research, we propose a new method to improve the weaknesses of the previous method. mCNN-ETC is a deep learning model which can transform the protein evolutionary information into image-like data composed of 20 channels, which correspond to the 20 amino acids in the protein sequence. We constructed CNN layers with different scanning windows in parallel to enhance the useful pattern detection ability of the proposed model. Then we filtered specific patterns through the 1-max pooling layer before inputting them into the prediction layer. This research attempts to solve a basic problem in biology in terms of application: predicting electron transporters and classifying their corresponding complexes. The performance result reached an accuracy of 97.41%, which was nearly 6% higher than its predecessor. We have also published a web server on http://bio219.bioinfo.yzu.edu.tw, which can be used for research purposes free of charge.

Key words: electron transport chain; five complexes; convolutional neural network; deep learning; position-specific scoring matrix; motif scanning

Introduction

Nature is an infinite source of inspiration for people to discover and recreate wonderful inventions. Inspired by how neurons work in the human brain [1, 2], the convolutional neural network (CNN) has been proposed and has become a powerful tool and is widely used in imaging-related tasks [3]. CNN and its structure variations increasingly evolved and reaped many state-of-the-art achievements not only in the field of computer vision but also in many other fields. In addition, CNN is also known as an effective tool in extracting hidden information in visual data. In bioinformatics, CNN has quickly received a lot of attention in the past decade, especially in biomedical images. However, the current solutions to apply CNN to nonvisual data (such as protein sequences) are still unresolved.

Protein function prediction has long been a significant and essential problem in the mission to investigate the role and influence of proteins *in vivo*. Because the cost of laboratory biological experiments is usually unaffordable, plus the time

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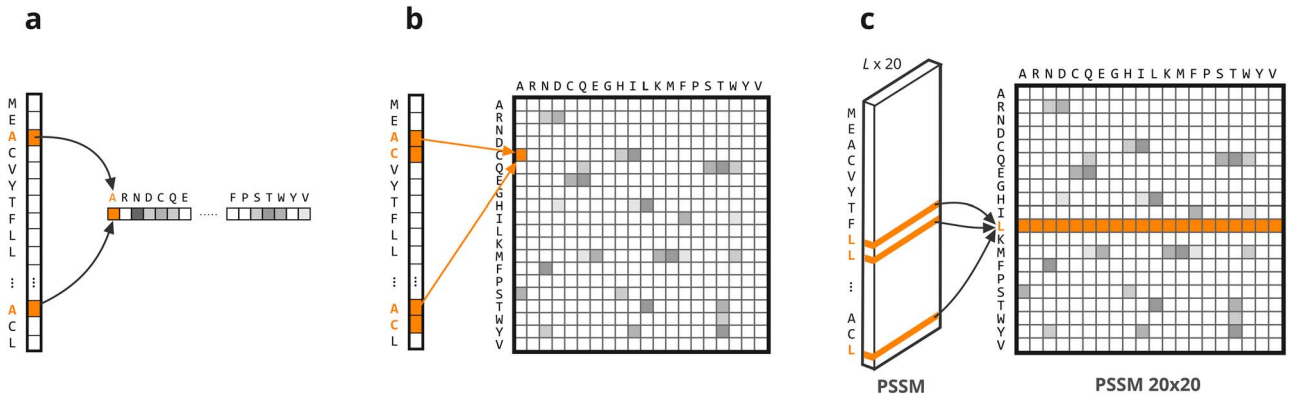


Figure 1. (A) Amino Acid Composition: Convert the protein sequence into a 20-element array containing the frequency of each amino acid in the sequence. (B) Dipeptide Composition: Convert the protein sequence to a 2D array by 20 × 20 containing the frequency of occurrence of each amino acid pair in the sequence. (C) Matrix PSSM 20 × 20: Similar to (B), the output after performed alignment with the PSI-BLAST tool will be summed up the same amino acids into 20 × 20 matrix, which is also the cause of the loss of position and order information in sequences.

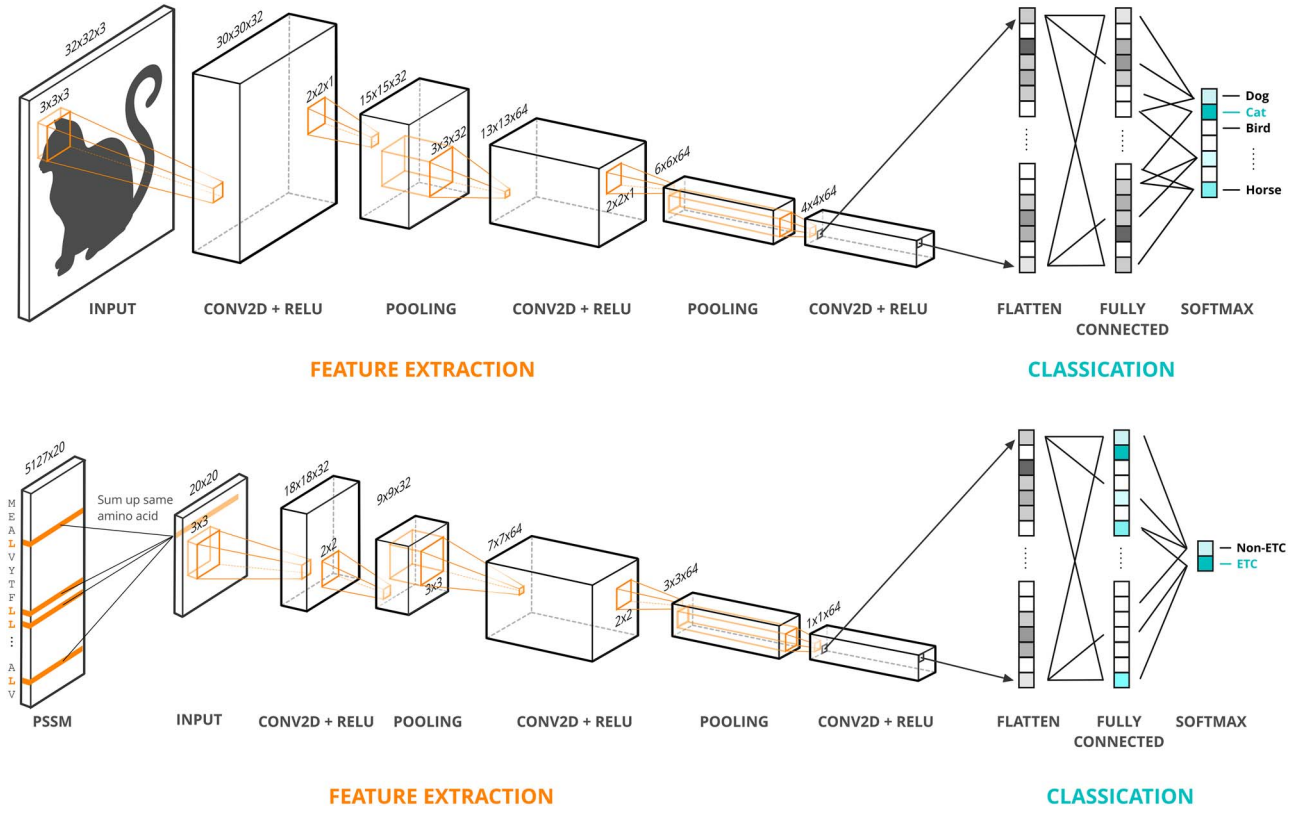


Figure 2. Structural comparison between the original CNN and the applied CNN using 20 × 20 PSSM matrix.

it takes, predicting the biological function of a protein based only on the primary sequence of the protein has become an exciting area that attracts researchers. Most efforts to solve this problem are usually divided into two categories: multiple sequences alignment-based and alignment-free-based.

In the first group, the structure and function of the unknown protein sequence can be predicted by its similarity with a large number of protein databases, which have been evaluated by multiple sequence alignment algorithms [4–6]. On the other hand, alignment-free-based methods use information based on primary sequence, such as the frequency or density of one or more amino acid pairs in the sequence (Figure 1A and B). The performance of alignment-free-based methods is generally inferior to that of multiple sequence alignment methods, making the methods based on multiple sequence alignments more interesting in the research community for a long time.

Unlike image data, protein sequences cannot be decomposed into color channels, and color channels provide a lot of useful information for CNN's pattern recognition capabilities. Another problem that makes it more troublesome to apply CNN to protein chains is the limitation of the input layer. The input layer is defined as a fixed dimension, and the length of the protein chain is usually variable. Many attempts have been made to implement CNN in protein prediction can be mentioned, such

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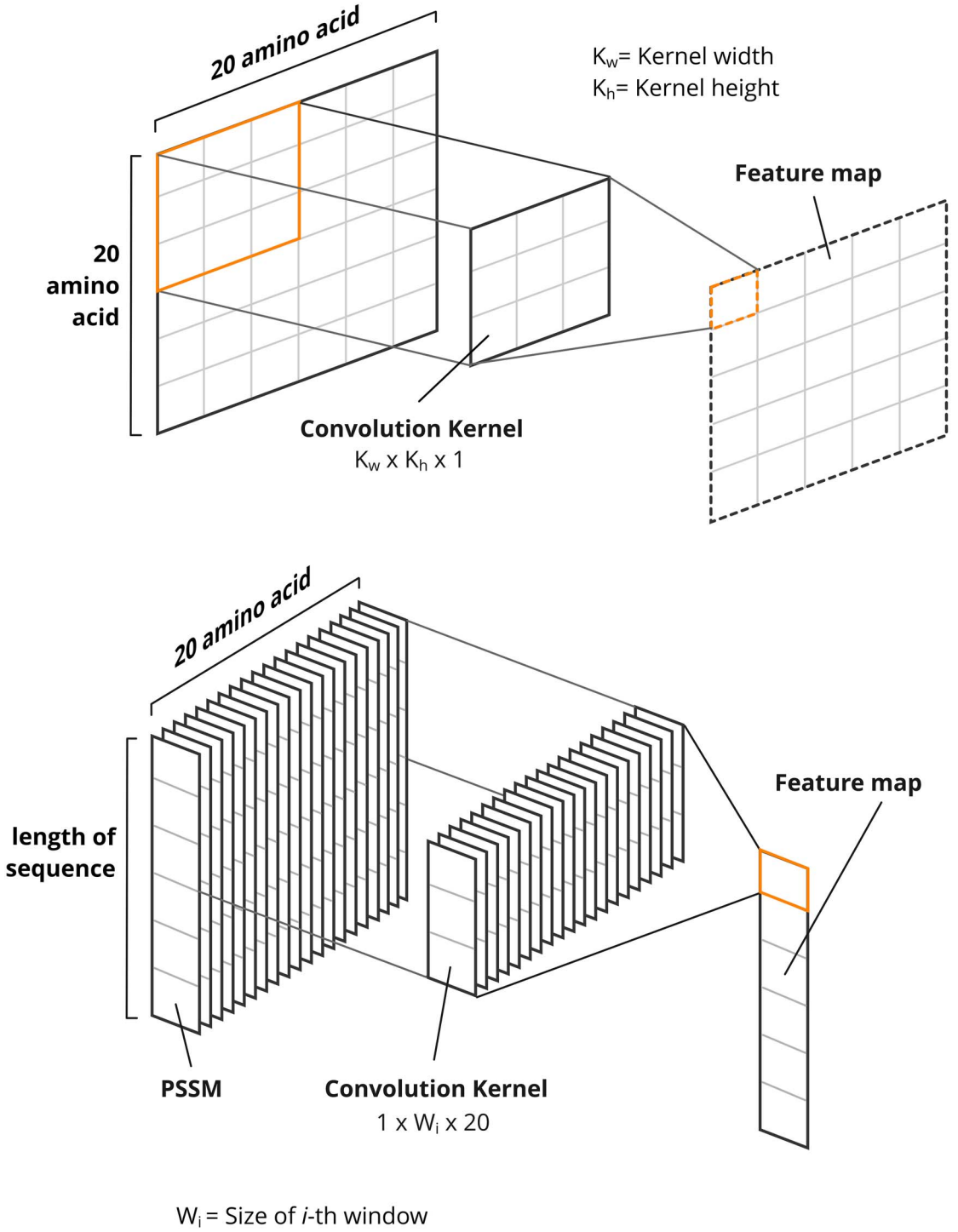


Figure 3. Input shape comparison: Instead of using the evolutionary information as 2D matrices, we transformed into 3D shape as similar as a long image with 20 channels. This helps CNN kernels scan more motifs that exist in the chain.

as the studies of Le et al. [7–9]. In this study, protein chains are converted into position-specific scoring matrices, which are also called evolutionary protein profiles, throughout an alignment tool. Similar to the idea from Amino Acid Composition (AAC), evolutionary protein profiles are then summed up with the same amino acids into a fixed 20×20 matrix before feeding them into CNN (Figure 1C). In this way, they become pioneers in implementing evolutionary information as a 2D input shape for CNN (Figure 2). However, in these studies, summing up evolutionary matrices is responsible for the loss of information about the position of amino acid elements in the chain. Hence, using this kind of improved information does not result in much improvement in prediction performance. In a recent study, Seo et al. [10] have proposed DeepFam, a deep learning model using multiple windows, to scan over binary matrices formed as a one-hot vector of a protein chain corresponding to 20 amino acids. Even though this study solved the difference in protein sequence lengths, the main weakness is the feature set. It is

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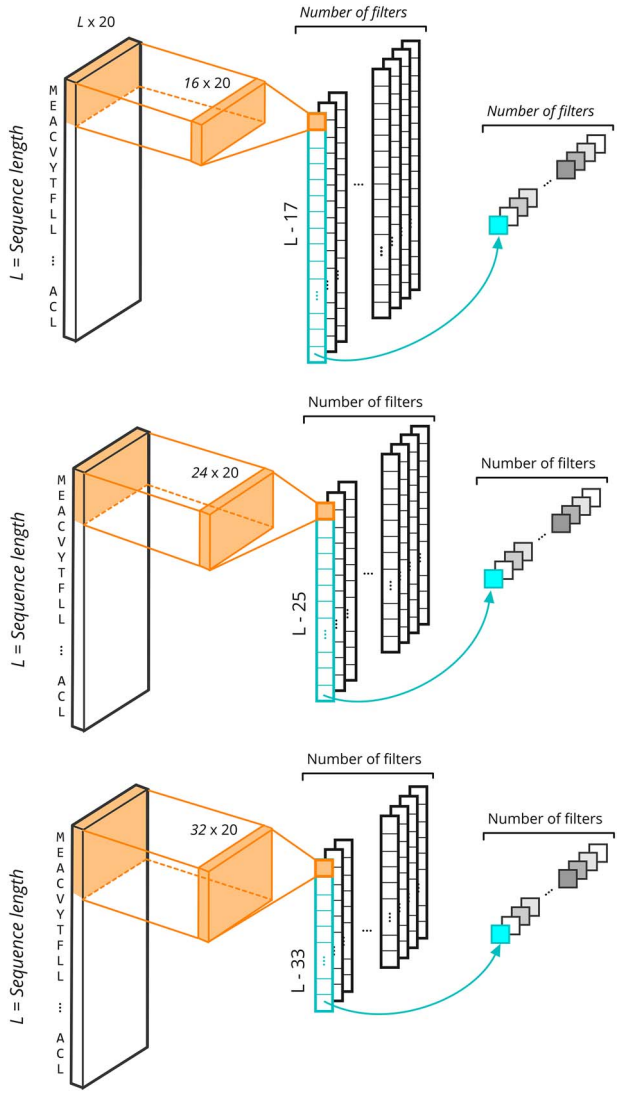


Figure 4. Multiple windows with size list = [16, 24, 32]. We constructed different window size for each convolutional layer to retain attentions which would be missed if only use single size. At the end of convolution layer, we add 1-max pooling layer to focus the significant motifs its outputs. Only one maximum value will be kept for each filter as a feature for the prediction step.

well known that although the protein sequences are made up of 20 essential amino acid units, these different positions do not share the same semantic meaning and distribute the same roles in the biological process. This story is similar to the problems in Natural language processing field (NLP). Ho et al. [11] anticipated the relationship between protein sequence and natural language and proposed an exciting strategy in their research and proposed a compelling method. They used well-trained models in NLP to interpret this ancient unknown language. In details, the study used the evolutionary information combine with data extracted from the linguistic knowledge vector space of the BERT [12] model to enhance the ability in protein identification. The critical role of multiple sequence alignment is also present in a famous publication recently. AlphaFold [13, 14] solved the protein folding problem with the help of a multiple sequence alignment technique before put it into their training process.

Here, we proposed an innovative method to apply a deep learning model using multiple window scanning techniques to solve a basic problem in biology: predicting electron transporters and classifying their corresponding complexes. In order to overcome the limitations compared with previous methods, we used the full evolutionary protein profiles as input data to maintain relevant data about the position and order of amino acids in the sequence. We kept the input shape the same as the image data, including 20 channels corresponding to 20 amino acids (Figure 3). In mCNN-ETC, we constructed CNN layers with different window sizes to scan the input data at the same time. At the end of each convolution layer, we used the 1-max pooling layer to hold the signature motif for each filter in different scanning windows (Figure 4).

We used a basic but important bioinformatics problem, predicting electron transporters and classifying their corresponding complexes, which has recently attracted many researchers [15, 16]. Figure 5 shows the flow of the electron transport chain in the inter-membrane space. In this process, there are five complexes: complex I, complex II, complex III, complex IV and complex V (also called ATP synthase). These complexes capture electrons from donor molecules and transfer them through biological processes. In the cell, each complex has a variety of electron carriers with different functions. Without the interaction of electron transport proteins, cells will not be able to obtain energy from the food they eat and affect all activities in our lives. Therefore, identifying electron transport proteins is essential to help biologists understand the electron transport chain and energy production in cells.

Table 1. Statistics of electron transport proteins in the survey dataset

	Original	Similarity <30%	Training	Independent
Electron transport proteins	2122	680	572	108
Transport proteins	10 206	3938	3274	664
Total	12 328	4618	3846	772

Table 2. Statistics of complexes in electron transport proteins in the survey dataset

	Original	Training	Independent
Complex I	84	63	21
Complex II	11	8	3
Complex III	20	15	5
Complex IV	18	14	4
Complex V	30	22	8
Unknown	517		
Total	680	122	41

Method

Data collection

First of all, we created a dataset in order to use as input to our proposed model. We collected sequences of transport proteins from UniProt [19, 20] using the clear query *'keyword:"Transport [KW-0813]' existence:"Evidence at protein level [1]" fragment:no AND reviewed:yes'* to ensure that only those proteins with annotation specified from experimental experiments. After this step, there are 12 328 sequences collected. We used BLAST [17] to exclude

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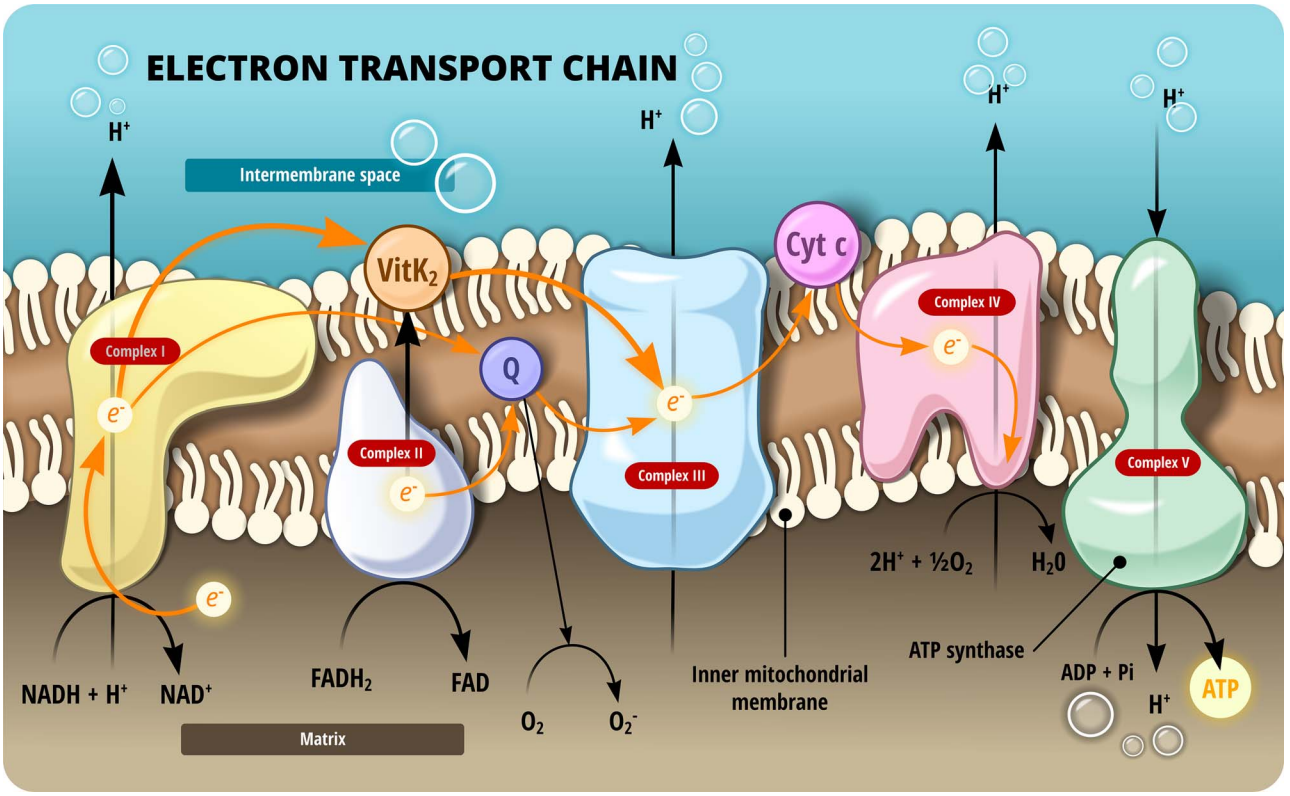


Figure 5. Cell respiration process with the aid of electron transport chain to keep and transfer electrons.

sequences with a sequence identity of more than 30% from the dataset and keep 4618 sequences in the next step. This refining step is necessary to prevent the overfitting problem in the training process caused by similarity in the whole survey dataset. Next, we labeled the electron transporters based on their explicit GO annotations (Supplementary Table 1) to obtain 772 electron transport proteins and 2846 for the rest class. At this point, we built two separated datasets corresponding to the two defined problems. For the electron transporter prediction, we randomly divided these sequences into the training and testing set. The training set contained 572 electron transporters and 3274 for the non-electron transporter class, while the independent set held 108 electron transport proteins and 664 general transporters. We carefully separated the dataset based on the complex labels for the second problem into extra training and testing sets for multiple classifications. Tables 1, 2 and Figure 6 list the statistics of our dataset in both problems.

Feature extraction

In this study, we prepared three feature sets replying to three approaches. As a first strategy, we compared the performance between the binary matrices used in DeepFam's research and the evolutionary protein profiles of the same survey dataset used in this study. Secondly, we built another feature set that is summed up with the same amino acid type used in Le et al.'s study. In the last step, we used the original evolutionary data directly from the BLAST tool's output to predict their functional families.

Firstly, we encoded protein sequences into binary matrices by using encoding follows IUPAC amino acid code notation, which is described by the following formula

$$X_{i,j} = \begin{cases} 1 & \text{if } s_i = j\text{th base in } charset \\ & \in \{A, C, D, E, F, G, H, I, K, L, M, N, P, Q, R, S, T, V, \\ & W, Y, X, O, U\} \\ 0.5 & \text{if } s_i = B \text{ and } j\text{th base in } charset \in \{D, N\} \\ & \text{else if } s_i = Z \text{ and } j\text{th base in } charset \in \{E, Q\} \\ & \text{else if } s_i = J \text{ and } j\text{th base in } charset \in \{I, L\} \\ 0 & otherwise \end{cases}$$

$$Y_i = \begin{cases} 1 & if\ ith\ protein\ is\ Electron\ Transport\ Protein \\ 0 & otherwise \end{cases}$$

where $i \in \{1, \ldots, L\}$. The output shape for this feature set is L x 21.

The protein evolutionary information or Position-Specific Scoring Matrix (PSSM) is a matrix of score values in positive or negative integers for representing motifs in biological sequences. They exploit multiple alignments and information related to protein families. In the profile, each amino acid position is reported by 20 values corresponding to 20 amino acids. It determines the frequency of substitution detected in a specific site in the protein family, of which negative values indicate that the specified amino acids are replaced less frequently and a positive value indicates that substitution occurs more frequently.

These well-known scoring matrices are widely used to predict protein secondary structure, subcellular locations and other biological information. We use the nonredundant protein database [18] together with the PSI-BLAST [19–21] tool in three iterators to convert sequences to PSSM. After collecting evolutionary protein profiles from conversion, we prepared the second feature set which simulated the experiment of Le et al.

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In this way, we were able to compare performance between different feature sets in the same deep learning model.

Multiple windows scanning

In this section, we focused on the description of *m*CNN-ETC. Inspired by DeepFam's study, we built a similar architecture consisting of various convolution layers with different window sizes to improve pattern recognition in the prediction task. The convolution layer scans the input throughout 20 channels and operates by windowing each convolution unit over the input sequence. Therefore, for the transformed sequence array of length L, the output size of the convolution layer is reduced to $L - W_k + 1$ for each W_k size of convolution unit, and we used ReLU activation function defined as:

$$\sigma(x) = \max(0, x) \tag{1}$$

At the end of each convolution layer, we also used 1-max pooling layer [22] to retain only the most important attention for each filter output.

$$h_k^{max} = \max_{i=1}^{l-m_k+1} (h_{k,i}) \tag{2}$$

Finally, we applied Dropout technique to avoid overfitting on the training process.

Assessment of predictive ability

We randomly divided the data set into two subsets: the independent testing set and the training set. In another hand, we also performed 5-fold cross-validation on the training set. We equally spitted into five parts, and we chose one of them as testing set, and the other four parts as training data. After five sequential tests, the cross-validation performance was calculated as the average of the five trials. We then used an independent set to test the model's stability and fairness in practice. We tried different hyper parameters including window sizes, number of filters and number of hidden nodes. The lengths of convolutional window sizes that we considered were 8, 12, 16, 20, 24, 28, 32 and 36, the number of filters was 256 and the number of hidden nodes in the fully connected layer was 1024. We also tried many combinations and chose the best parameter set for our final model. The best hyperparameters were set as follows: coefficient of regularization (λ) =0.0005, dropout rate = 30%, learning rate = 0.001, batch size =128.

Sensitivity, specificity, accuracy and Matthew's correlation coefficient (MCC) was used to measure the prediction performance. TP, FP, TN and FN are true positives, false positives, true negatives and false negatives, respectively.

$$Sensitivity = \frac{TP}{TP + FN} \tag{3}$$

$$Specificity = \frac{TN}{TN + FP} \tag{4}$$

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{5}$$

$$MCC = \frac{TP * TN - FP * FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{6}$$

Result

Analysis of the architecture and the performance of *m*CNN-ETC

In this section, we performed multiple alignments to convert protein sequence to the evolutionary information, which holds helpful scoring about each amino acid's position in its chain. To solve the problem of difference in sequence lengths, we then applied the zero-padding to bring all the evolution matrices into the same dimension by Lx20, where L is the maximum of all sequences. For the mCNN-ETC, we have defined three independent arrays to construct motif extraction layers consisting of a convolution layer of kernel size is 1xm_i, where m_i is the length of the i-th in the window list. Instead of using the input data as a 2D matrix by Lx20, we transformed the input matrix into a three-dimensional matrix by 1xLx20 so that CNN kernels could consider the input as an image with 20 channels and try to scan for features similar to the way they did with visual data.

In the first stage, we performed training the electron prediction dataset with individual convolution sizes to discover how much the difference in performance and recognition power of each single convolution length on the protein sequence. Prediction performance with different kernel lengths (Supplementary Table 2) shows a noticeable improvement in accuracy when increasing the length of the kernel size. Along with that, the positive class recognition ability also increases among the length of the sliding window. We then experimented with different numbers of filters and hidden nodes to discover whether the prediction performance depends on the number of these hyperparameters. The results (Supplementary Tables 3 and 4) also indicated that the electron transport prediction problem with modest data would give similar results. We experimented on several different size combinations and picked the one with the most promising performance in the next step. The entire implementation for the electron transport prediction problem is shown in Figure 7.

To make a comparison with previous methods, we also reconstructed the CNN model which described details in the Le's paper. The difference in performance displayed in Table 3 explains that, although only the fundamental binary matrix is used, the model with various length sizes provides better motif detection and good prediction capability. Compared to the one-hot matrix, PSSM 20x20's performance is almost the same; however, the sensitivity was slightly higher. The performance of our proposed method provided 6% higher accuracy and achieved a balance between sensitivity and specificity. We also generated the area under the curve (AUC) and receiver operating characteristics (ROC) curves in Figures 8a and b as evaluation metrics for checking how much our models are capable of distinguishing between classes. With a higher AUC value, the proposed method has been confirmed to excel in predicting electron transport proteins in general proteins. We also trained another model with the same structure on the final stage of the prediction step using the rest dataset for predicting complex tasks. Table 4 shows high results including 92,68%, 97,56%, 97,56%, 100% and 97,56% in the complex predicting problem, respectively.

Visualize significant motif inside mCNN-ETC

In order to better understand and find the most specific motifs for the survey dataset, we first decomposed the well-trained mCNN-ETC and collected only the 1-max pooling layers' outputs.

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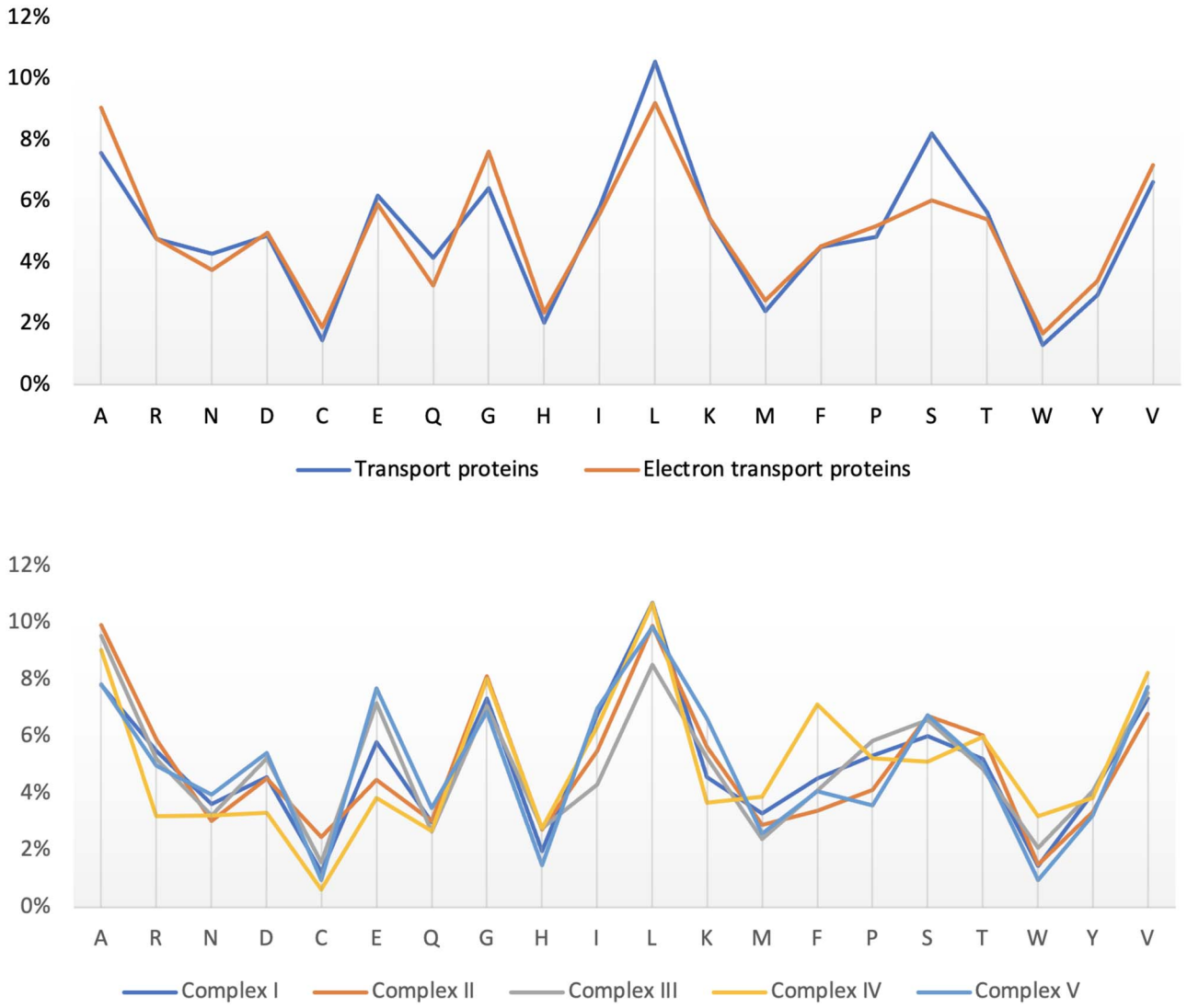


Figure 6. Amino acid composition of electron transport proteins and transport proteins and five complexes in the survey dataset.

Table 3. Prediction performance with different feature sets

Feature sets	Sens	Spec	Acc	MCC
Cross-validation (5-folds)				
Binary matrix	51.49% ± 8.46%	97.50% ± 0.72%	92.21% ± 2.21%	0.5763 ± 0.0790
Old method	45.43% ± 12.04%	98.88% ± 0.55%	92.41% ± 1.93%	0.5853 ± 0.0784
Proposed method	77.52% ± 5.74%	98.78% ± 0.59%	95.97% ± 1.34%	0.8171 ± 0.0468
Independent				
Binary matrix	59.26%	97.29%	91.97%	0.6366
Old method	65.74%	95.78%	91.58%	0.6383
Proposed method	92.59%	98.19%	97.41%	0.8942

We used the Random Forest classifier, an ensemble learning method fundamentally based on decision trees, to rank all archived filters from the independent set. We then selected the top ranking of the filter to perform the visualization. In the next step, we took several electron transport proteins in the testing set and put them into the model for performing prediction, and measured the position where convolution kernels detected the most hopeful motif on the evolutionary information. Figure 9 shows the correlation between the evolutionary motif and the corresponding filter. We can easily see the correlation between the corresponding motif and the filter, which focuses on the amino acid 'C' in all three sample proteins observed in the same way that an original CNN model could detect the object's features in the image data. In our proposed model, motifs

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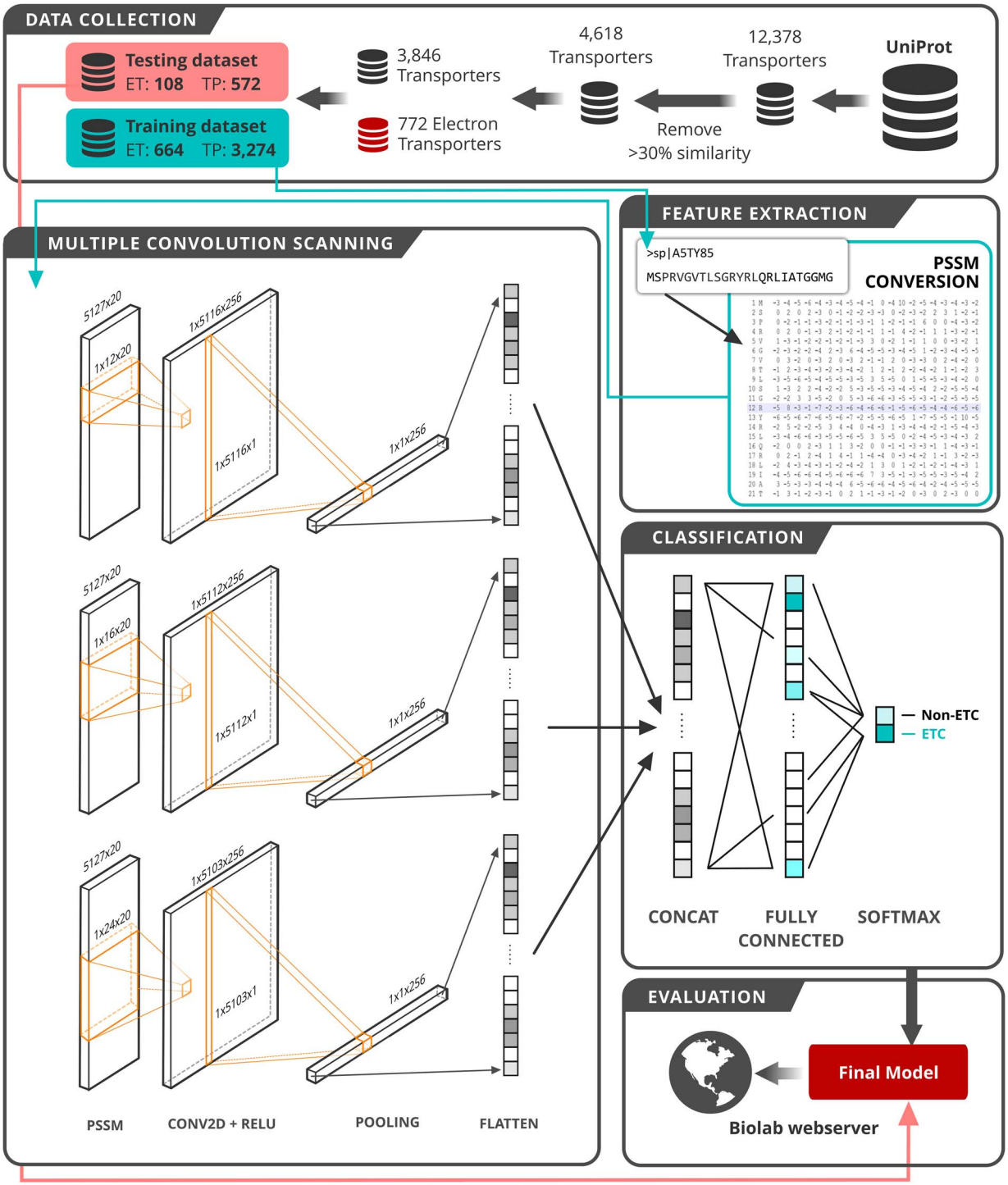


Figure 7. The workflow for building electron protein prediction model.

were explored and saved by multiple windows instead of a single window. It made pattern recognition more efficient, without missing motifs with different sizes throughout the length of the protein sequence. In each of the instances given, we can easily observe that the similarity occurs in specific regions, even on different proteins. In general, the protein recognition performance has been improved due to applying multiple length kernel scans on the protein evolutionary information.

Web service of mCNN-ETC

Based on the achieved results, we confidently built a webserver to demonstrate our research. *m*CNN-ETC is trained to identify

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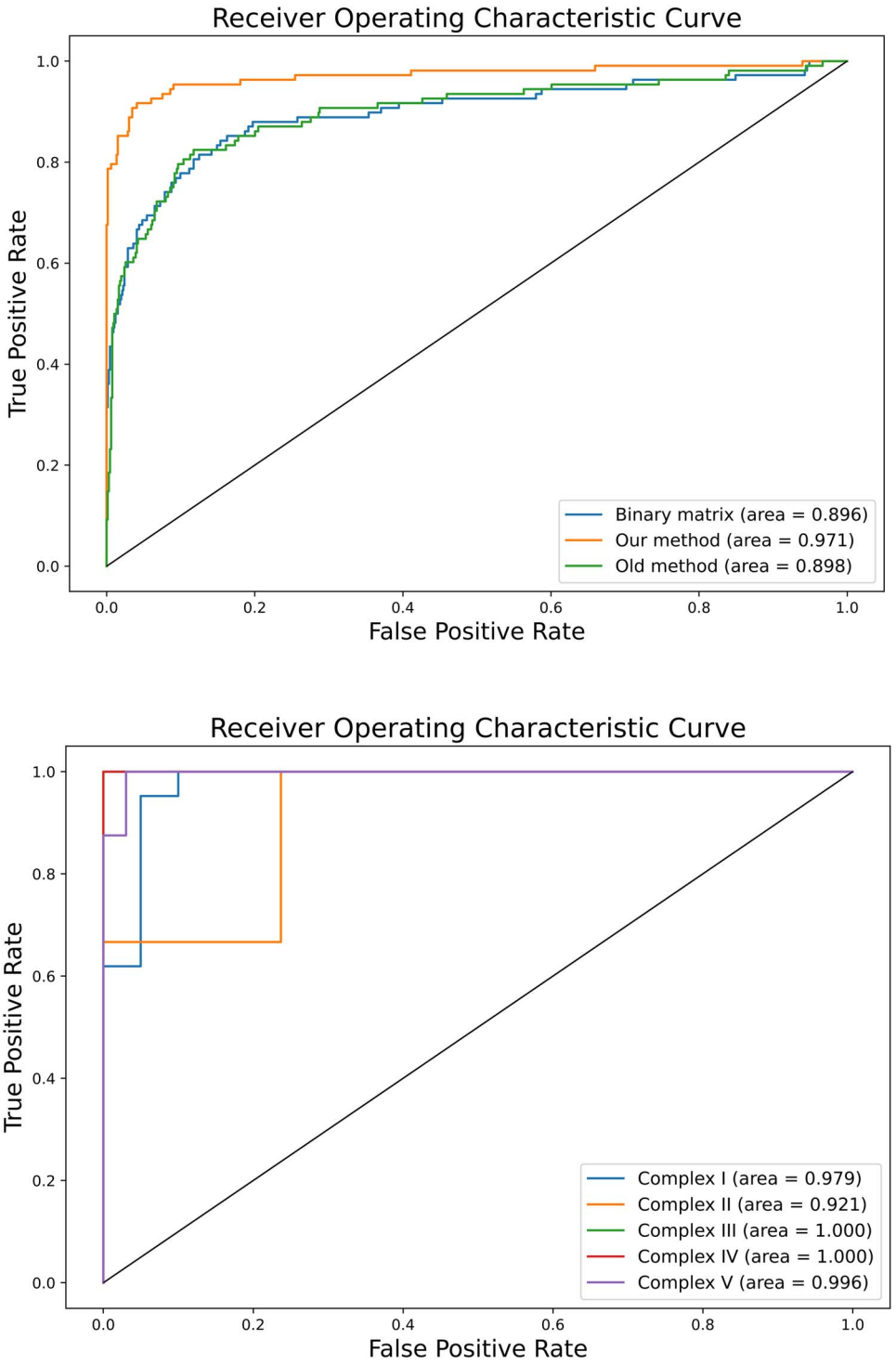


Figure 8. (a) ROC curve for performance of predicting electron transport proteins with different feature sets. (b) ROC curve for performance of predicting different complexes.

electron transport proteins in transporters and identify their complexes. We have implemented a user-friendly interface that is ready for users to submit their sequences and receive information later via the results page and email notification. To keep the webserver up and running reliably, we have applied queues for predictive operations because these tasks consume computing resources and execution time. *m*CNN-ETC can be accessed at http://bio219.bioinfo.yzu.edu.tw. Through this webserver, we hope to contribute our research to the community and help biologists explore electron transport protein in transporters to understand better the mechanism of electron transport in the transport chain.

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Table 4. Prediction performance electron transporter chains' complexes

	TP	FP	TN	FN	Sens	Spec	Acc	MCC
Complex I	21	3	17	0	100.00%	85.00%	92.68%	0.8624
Complex II	2	0	38	1	66.67%	100.00%	97.56%	0.8060
Complex III	4	0	36	1	80.00%	100.00%	97.56%	0.8823
Complex IV	4	0	37	0	100.00%	100.00%	100.00%	1.0000
Complex V	7	0	33	1	87.50%	100.00%	97.56%	0.9216

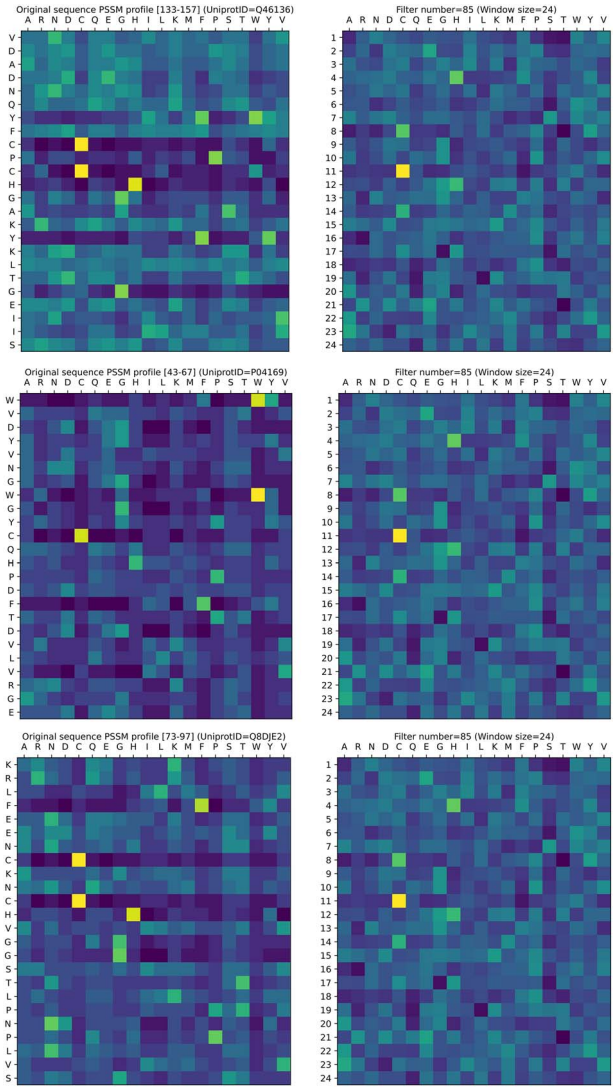


Figure 9. Visualize the significant motif and its corresponding position in chain. The most characteristic motif visualization is found (on the right) through the importance ranking using Forest Ensemble in the electron transport protein prediction. Brighter colors represent high scoring values in evolutionary information.

Conclusion

Although more than a decade ago, CNN has never ceased to attract scientists' attention for its powerful ability to identify hidden features in data. This study proposed a method using CNN with protein chains in the natural way that CNN did with imaging. We demonstrated performance in two problems of predicting electron transport in general proteins and then identifying their complexes. Compared to previous studies, we retained position and order information in chains, and at the same time, also distilled the proposed model to examine the binding motifs in protein sequences. This study only focuses on developing a deep learning model to recognize motifs in electron transport chains through evolutionary information. Hence, we did implement to demonstrate detecting ability in a fundamental but necessary problem. Based on the state-of-the-art results in this study, we believe that more problems in bioinformatics will be resolved effectively, such as binding site identifying, protein structure or even protein interaction prediction. Moreover, we proposed our CNN framework with multiple windows scanning and protein evolution information may also be a good tool for other protein classification problems. We have built the web service of our proposed method. Then we opened our source code for public research.

Key Points

- Using a Deep Learning model with multiple windows in scanning the protein evolutionary information.
- Use a particular input method such as an image with 20 channels, keep the order and position information from the protein sequence.
- Visualize the most significant pattern in electron transport prediction. Reply on the state-of-the-art performance compared to previous works.

Supplementary data

Supplementary data are available online at *Briefings in Bioinformatics*.

Authors' contributions

Q.T.H., N.Q.K.L. and Y.Y.O. conceived the idea. N.Q.K.L. and Q.T.H. collected dataset. Q.T.H. developed the programs, Q.T.H. performed the writing and original draft preparation. Q.T.H. and N.Q.K.L. did the writing, review and editing. Q.T.H., N.Q.K.L. and Y.Y.O. oversaw the project. N.Q.K.L. and Y.Y.O. acquired the funding. Y.Y.O. read and agreed to the published version of the manuscript.

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Competing Interests

The authors declare that they have no competing interests.

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References

1. Hubel DH, Wiesel TN. Receptive fields and functional architecture of monkey striate cortex. *J Physiol* 1968;**195**:215–43.
2. Fukushima K. Neocognitron: a self-organizing neural network model for a mechanism of pattern recognition unaffected by shift in position. *Biol Cybern* 1980;**36**:193–202.
3. LeCun Y, Boser B, Denker JS, *et al*. Backpropagation applied to handwritten zip code recognition. *Neural Comput* 1989;**1**:541–51.
4. Thompson JD, Higgins DG, Gibson TJ. CLUSTAL W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. *Nucleic Acids Res* 1994;**22**:4673–80.
5. Sievers F, Wilm A, Dineen D, *et al*. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal omega. *Mol Syst Biol* 2011;**7**:539.
6. Edgar RC. MUSCLE: multiple sequence alignment with high accuracy and high throughput. *Nucleic Acids Res* 2004;**32**:1792–7.
7. Le N-Q, Ho Q-T, Ou Y-Y. Incorporating deep learning with convolutional neural networks and position specific scoring matrices for identifying electron transport proteins. *J Comput Chem* 2017;**38**:2000–6.
8. Le N-Q-K, Ho Q-T, Ou Y-Y. Classifying the molecular functions of Rab GTPases in membrane trafficking using deep convolutional neural networks. *Anal Biochem* 2018;**555**:33–41.
9. Le NQK, Ho Q-T, Yapp EKY, *et al*. DeepETC: a deep convolutional neural network architecture for investigating and classifying electron transport chain's complexes. *Neurocomputing* 2020;**375**:71–9.
10. Seo S, Oh M, Park Y, *et al*. DeepFam: deep learning based alignment-free method for protein family modeling and prediction. *Bioinformatics* 2018;**34**:i254–62.
11. Ho Q-T, Nguyen T-T-D, Khanh Le NQ, *et al*. FAD-BERT: improved prediction of FAD binding sites using pre-training of deep bidirectional transformers. *Comput Biol Med* 2021;**131**:104258.
12. Devlin J, Chang M-W, Lee K, *et al*. BERT: pre-training of deep bidirectional transformers for language understanding. arXiv preprint arXiv:1810.04805, 2018.
13. Senior AW, Evans R, Jumper J, *et al*. Improved protein structure prediction using potentials from deep learning. *Nature* 2020;**577**:706–10.
14. Jumper J, Evans R, Pritzel A, *et al*. Highly accurate protein structure prediction with AlphaFold. *Nature* 2021. https://doi.org/10.1038/s41586-021-03819-2.
15. Bajpai R, Sharma A, Achreja A, *et al*. Electron transport chain activity is a predictor and target for venetoclax sensitivity in multiple myeloma. *Nat Commun* 2020;**11**:1228.
16. Nguyen D, Ho-Quang T, Dinh-Phan V, *et al*. Use Chou's 5-steps rule with different word embedding types to boost performance of electron transport protein prediction model. In *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2020. doi: 10.1109/TCBB.2020.3010975.
17. Tatusova TA, Madden TL. BLAST 2 sequences, a new tool for comparing protein and nucleotide sequences. *FEMS Microbiol Lett* 1999;**174**:247–50.
18. Pruitt KD, Tatusova T, Maglott DR. NCBI reference sequences (RefSeq): a curated non-redundant sequence database of genomes, transcripts and proteins. *Nucleic Acids Res* 2007;**35**:D61–5.
19. Altschul SF, Koonin EV. Iterated profile searches with PSI-BLAST—a tool for discovery in protein databases. *Trends Biochem Sci* 1998;**23**:444–7.
20. Altschul SF, Madden TL, Schäffer AA, *et al*. Gapped BLAST and PSI-BLAST: a new generation of protein database search programs. *Nucleic Acids Res* 1997;**25**:3389–402.
21. Schäffer AA, Aravind L, Madden TL, *et al*. Improving the accuracy of PSI-BLAST protein database searches with composition-based statistics and other refinements. *Nucleic Acids Res* 2001;**29**:2994–3005.
22. Alipanahi B, Delong A, Weirauch MT, *et al*. Predicting the sequence specificities of DNA- and RNA-binding proteins by deep learning. *Nat Biotechnol* 2015;**33**:831–8.

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